# Circular Measure

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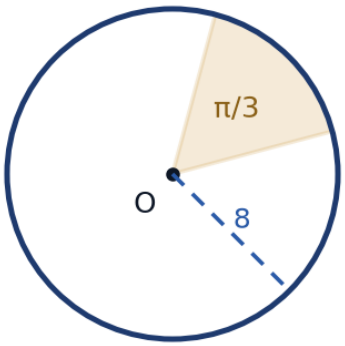
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Converting between degrees and radians**

Convert $135^\circ$ to radians, in terms of $\pi$.

**2** **[Foundational]** • **Target: Arc length**

A circle has radius 8 cm. Find the length of an arc subtending an angle of $\frac{\pi}{3}$ radians at the center, using $s = r\theta$.




**3** **[Foundational]** • **Target: Sector area**

Find the area of a sector with radius 6 cm and angle $\frac{2\pi}{3}$ radians, using $A = \frac{1}{2}r^2\theta$.

**4** **[Foundational]** • **Target: Sector area**

A sector has radius 10 cm and area $40$ cm$^2$. Find the angle $\theta$ in radians.

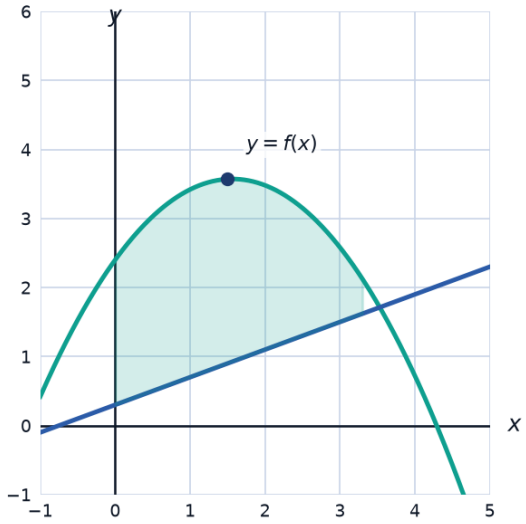
**5** **[Foundational]** • **Target: Perimeter of a sector**

Find the perimeter of a sector with radius 7 cm and angle 1.2 radians.

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**6** [Foundational] • Target: The area of a segment

A circle has radius 5 cm. A chord subtends an angle of $\frac{\pi}{2}$ radians at the center. Find the area of the minor segment, using segment area = sector area minus triangle area.




**7** [Foundational] • Target: Small-angle approximations

Use the small-angle approximations $\sin\theta \approx \theta$ and $\cos\theta \approx 1 - \frac{\theta^2}{2}$ to approximate $\sin(0.1)$ and $\cos(0.1)$, to four decimal places.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Small-angle approximations

Use the small-angle approximation $\tan\theta \approx \theta$ to approximate $\tan(0.2)$, and compare it to the exact value $\tan(0.2) \approx 0.2027$.

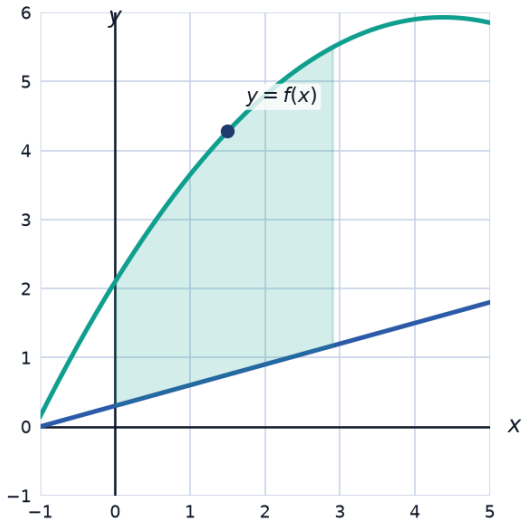
**9** [Intermediate] • Target: Angular speed and radians per second

A wheel rotates at 5 radians per second. Find the angle turned through, in radians, after 8 seconds, and convert this to the number of complete revolutions.

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**10** **[Intermediate]** • **Target: A well-designed word problem: a swinging pendulum's arc**

This question has two parts. A pendulum of length 1.2 meters swings through an angle of $0.3$ radians on each side of vertical (a total swing angle of $0.6$ radians). (a) Find the length of the arc traced by the pendulum's tip during one full swing (from one side to the other). (b) Find the area swept out by the pendulum during this swing.




**11** **[Intermediate]** • **Target: Sector area**

For the ellipse $\frac{x^2}{81} + \frac{y^2}{9} = 1$, find the length of the major axis and the foci.

**12** **[Intermediate]** • **Target: Perimeter of a sector**

$x1200isinvestedatasimpleinterestrateof8\%$ per year for 5 years. Find the interest earned.

**13** **[Intermediate]** • **Target: The area of a segment**

How many distinct arrangements are there of the letters in the word LEVEL? Show your complete algebraic work.

**14** **[Intermediate]** • **Target: Small-angle approximations**

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches $1.5\times$ its initial size (i.e. $P = 1.5 \cdot 500$)? Show your complete algebraic work.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

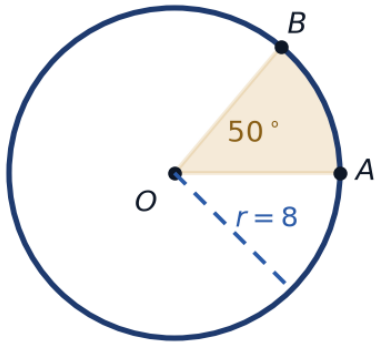
**15** **[Challenging]** • **Target: Angular speed and radians per second** • **Cross-Lesson: 5.01 Trigonometry**

Find the area of the polar region bounded by $r = 2$ for $0 \le \theta \le \frac{\pi}{6}$.

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**16** [Challenging] • Target: A well-designed word problem: a swinging pendulum's arc • Cross-Lesson: 6.01 Differentiation & Integration

Find the reference angle for $220^\circ$.




**17** [Challenging] • Target: Converting between degrees and radians • Cross-Lesson: 7.01 Statistics & Probability

Solve $\cos(\theta) = -\frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the smallest solution (in degrees).

**18** [Challenging] • Target: Arc length • Cross-Lesson: 8.01 Functions: Composite & Inverse

A right triangle has one leg of length 12 and hypotenuse 20. Find the length of the other leg.

**19** [Challenging] • Target: Sector area • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Solve $\sin(\theta) = \frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the largest solution (in degrees).

**20** [Challenging] • Target: Perimeter of a sector • Cross-Lesson: 10.01 Series & Binomial Theorem

In $\Delta ABC$, $a = 5$, $b = 7$, and $\angle C = 60^\circ$. Find side $c$ (Law of Cosines).

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Converting between degrees and radians

A sector of a circle of radius $12$ cm has an arc length of $18$ cm. Find the area of the sector.

**a.** $108\ \text{cm}^2$

**b.** $216\ \text{cm}^2$

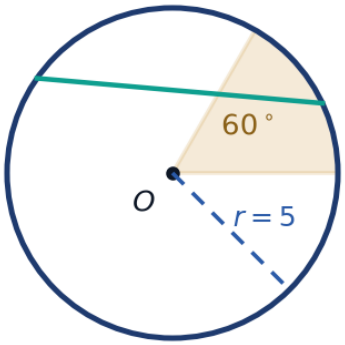
**c.** $54\ \text{cm}^2$

**d.** $144\ \text{cm}^2$

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**22** [Foundational] • Target: Arc length

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?




**a.** 6 units

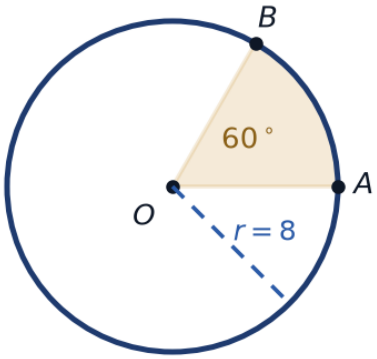
**b.** $6\sqrt{3}$ units

**c.** 12 units

**d.** $12\sqrt{3}$ units

**23** [Foundational] • Target: Sector area

A circle has radius 8 cm. Find the arc length subtended by an angle of 1.5 radians.




**a.** 12 cm

**b.** 6 cm

**c.** 24 cm

**d.** 9.6 cm

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**24** [Foundational] • Target: Perimeter of a sector

A sector's perimeter is 26 cm and its radius is 8 cm. Find the angle of the sector, in radians.

**a.** 1.25

**b.** 1.625

**c.** 0.8

**d.** 2

**25** [Foundational] • Target: The area of a segment

A sector of a circle has arc length $5\pi$ cm and angle $180^\circ$. Find the radius.

**a.** 1

**b.** 5

**c.** 2

**d.** 3

**26** [Foundational] • Target: Small-angle approximations

A sector of a circle has radius $9$ cm and angle $45^\circ$. Find the exact arc length.

**a.** $\frac{5\pi}{4}$

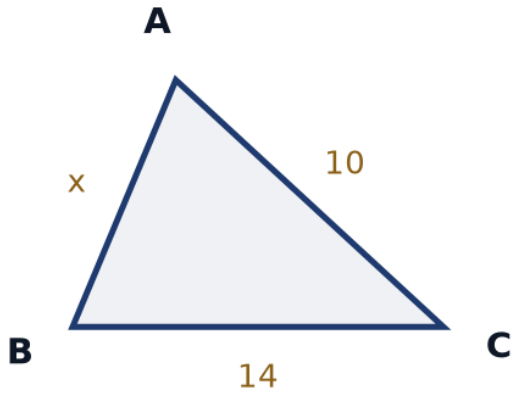
**b.** $\frac{9\pi}{4}$

**c.** $\frac{3\pi}{2}$

**d.** $\frac{7\pi}{4}$

**27** [Foundational] • Target: Angular speed and radians per second

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?




**a.** 10.3

**b.** 12.1

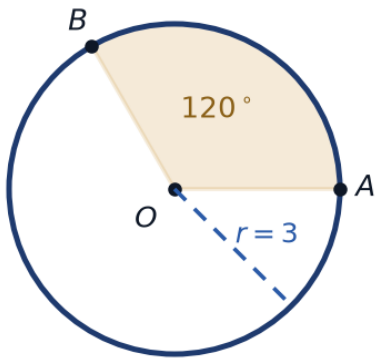
**c.** 14.8

**d.** 8.5

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**28** [Foundational] • Target: A well-designed word problem: a swinging pendulum's arc

The function $f$ is defined for all real numbers $x$ by $f(x) = 2 - 3\sin(4x)$. What are the amplitude and the period (in radians) of $f$?




**a.** Amplitude $= 3$, Period $= \frac{\pi}{2}$

**b.** Amplitude $= -3$, Period $= \frac{\pi}{2}$

**c.** Amplitude $= 3$, Period $= 8\pi$

**d.** Amplitude $= 5$, Period $= \frac{\pi}{4}$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Converting between degrees and radians

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60^\circ$, find the length of side $c$. Round your answer to two decimal places.

**a.** 13.23

**b.** 14.70

**c.** 12.96

**d.** 15.52

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**30** [Intermediate] • Target: Arc length

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

**a.** $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$

**b.** $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

**c.** $x = \frac{\pi}{6},\ \frac{7\pi}{6}$

**d.** $x = 0,\ \pi$

**31** [Intermediate] • Target: Sector area

Given $\mathbf{a} = (2,\ -6,\ -9)$ and $\mathbf{b} = (2, 3, 6)$, find $\cos\theta$, where $\theta$ is the angle between them.

**a.** $\frac{9}{77}$

**b.** $\frac{68}{77}$

**c.** $-\frac{34}{9}$

**d.** $-\frac{68}{77}$

**32** [Intermediate] • Target: Perimeter of a sector

Find the angle between $\mathbf{u} = \langle 1, 0\rangle$ and $\mathbf{v} = \langle 0, 1\rangle$.

**a.** $135^\circ$

**b.** $60^\circ$

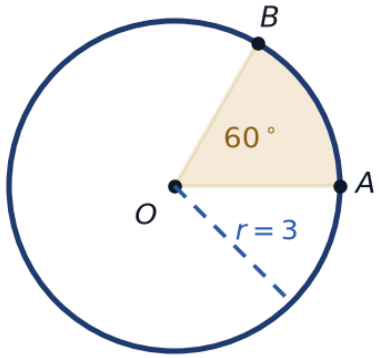
**c.** $90^\circ$

**d.** $120^\circ$

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**33** **[Intermediate]** • **Target: The area of a segment**

A particle moves in the xy-plane. Its velocity vector $\mathbf{v}(t)$ at time $t$ is given by $\mathbf{v}(t) = \langle 3t^2 - 6t, e^t - 1 \rangle$, where $t \geq 0$. The initial position of the particle at $t = 0$ is $\mathbf{r}(0) = \langle 5, -2 \rangle$. Find the position vector $\mathbf{r}(t)$ of the particle at time $t$. Then, determine the value of $t$ for which the particle's speed is minimized and state the minimum speed.




**a.** $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{2}$ at $t = 1$

**b.** $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{5}$ at $t = 0$

**c.** $\mathbf{r}(t) = \langle 3t^2 - 6t + 5, e^t - t - 2 \rangle$, minimum speed $0$ at $t = 0$

**d.** $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{3}$ at $t = 0$

**34** **[Intermediate]** • **Target: Small-angle approximations**

The first, fourth, and sixth terms of an arithmetic progression $(u_n)$ are equal to the first, second, and third terms, respectively, of an infinite geometric progression $(v_n)$ with common ratio $re1$. Given that the sum to infinity of the geometric progression is $27$, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first $n$ terms of the arithmetic progression.

**a.** 44

**b.** 45

**c.** 50

**d.** 54

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**35** [Intermediate] • Target: Angular speed and radians per second

Let the complex number $z_1$ be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, $z_2$, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of $\theta_2$ in radians, given that $0 \le \theta_2 < 2\pi$.

**a.** $\frac{\pi}{2}$

**b.** $\frac{2\pi}{3}$

**c.** $\frac{5\pi}{6}$

**d.** $\frac{\pi}{6}$

**36** [Intermediate] • Target: A well-designed word problem: a swinging pendulum's arc

Consider a geometric progression where the sum of the first $n$ terms is given by $S_n = \frac{3(2^n - 1)}{2}$. A second sequence, denoted by $a_n$, is defined such that $a_n = S_n - S_{n-1}$ for $n \ge 2$, and $a_1 = S_1$. Find the value of $a_5 - a_3$.

**a.** 24

**b.** 18

**c.** 48

**d.** 36

**37** [Intermediate] • Target: Converting between degrees and radians

The continuous function $f$ is defined for all $x \in \mathbb{R}$ by $f(x) = 2^x + 3x - 11$. Between which pair of consecutive integers does a root of the equation $f(x) = 0$ lie?

**a.** 2 and 3

**b.** 1 and 2

**c.** 3 and 4

**d.** 0 and 1

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Arc length • Cross-Lesson: 8.01 Functions: Composite & Inverse

Relative to a fixed origin $O$, the points $A$ and $B$ have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.

**a.** 7

**b.** $\sqrt{31}$

**c.** $\sqrt{69}$

**d.** 49

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**39** [Challenging] • Target: Sector area • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The function $f$ is defined for all real numbers by $f(x) = x^3 + 3x - 5$. Between which pair of consecutive integers does the unique real root of the equation $f(x) = 0$ lie?

**a.** $1$ and $2$

**b.** $0$ and $1$

**c.** $2$ and $3$

**d.** $-1$ and $0$

**40** [Challenging] • Target: Perimeter of a sector • Cross-Lesson: 10.01 Series & Binomial Theorem

A car purchased for $x20000depreciatesatarateof20\%$ per year (reducing balance). Find its value after 3 years, correct to 2 decimal places.

**a.** 10240.00

**b.** 9640.00

**c.** 10540.00

**d.** 9940.00

**41** [Challenging] • Target: The area of a segment • Cross-Lesson: 11.01 Kinematics

$x1500isinvestedatacompoundinterestrateof7\%$ per year, compounded annually, for 5 years. Find the value of the investment, correct to 2 decimal places.

**a.** 2103.83

**b.** 2063.83

**c.** 2143.83

**d.** 2023.83

**42** [Challenging] • Target: Small-angle approximations • Cross-Lesson: 12.01 Forces & Equilibrium

Find a positive coterminal angle with $400^\circ$ in $[0^\circ, 360^\circ)$.

**a.** 30

**b.** 40

**c.** 31

**d.** 400

**43** [Challenging] • Target: Angular speed and radians per second • Cross-Lesson: 13.01 Newton's Laws

Given $z = r(\cos 90^\circ + i\sin 90^\circ)$, find the principal argument (in degrees, $0 \le \theta < 360$) of $z^3$.

**a.** 270

**b.** 0

**c.** 274

**d.** 90

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**44** [Challenging] • Target: A well-designed word problem: a swinging pendulum's arc • Cross-Lesson: 14.01 Energy, Work & Power

Find the reference angle of $120^\circ$.

**a.** 120

**b.** 60

**c.** 31

**d.** 30

**45** [Challenging] • Target: Converting between degrees and radians • Cross-Lesson: 15.01 Representation of Data

Find the area enclosed by the curve $y = x^2$, the x-axis, and the line $x = 3$, for $0 \le x \le 3$.

**a.** 9

**b.** 5

**c.** 4

**d.** 6

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